

Mashrur Mahmood

+880-1911-583682 | mashrur.mahmood@gmail.com | linkedin.com/in/mashrur-mahmood
github.com/mashmahmood | codeforces.com/profile/iamashrur

SUMMARY

Persevering and analytical, with strong problem-solving skills developed through multidisciplinary engineering projects, competitive programming, and software development. Thrives in collaborative environments, values continuous learning, and is committed to delivering reliable, innovative solutions to real-world engineering challenges.

EDUCATION

Bangladesh University of Engineering and Technology

Bachelor of Science in Mechanical Engineering

Dhaka, Bangladesh

Nov 2021 – Jun 2026

- **CGPA:** 3.67/ 4.00
- **Awards:** University Merit Stipend (3 Terms)

Saint Joseph Higher Secondary School

Higher Secondary Certificate, Science Division

Dhaka, Bangladesh

2020

- **GPA:** 5.00/ 5.00
- **Awards:** Perfect Attendance

EXPERIENCE

Industrial Attachment

United Ashuganj Energy Ltd. (UAEL) | [Report \(PDF\)](#)

March 2025

Ashuganj, Bangladesh

- Completed a 13-day industrial training program at a 195 MW combined-cycle gas-engine power plant.
- Studied the operation of gas engines, HRSGs, steam turbines, gas RMS, substations, and auxiliary mechanical and electrical systems.
- Acquired practical exposure to DCS/SCADA operations, plant safety, preventive maintenance, and utility-scale power plant operation.

PROJECTS

Autonomous Indoor UAV SLAM and Thermal Surveying

Undergraduate Project and Thesis

Apr 2025 – Jun 2026

[Book \(PDF\)](#) | [Demo \(Video\)](#)

- Designed, assembled, and integrated an autonomous quadcopter for GPS-denied indoor navigation.
- Key Components: Raspberry Pi, Pixhawk 6C, Optical Flow Sensor, Distance Sensor, LiDAR, and Thermal Imager
- Frameworks & Libraries: PX4, ROS2, Gazebo, slam-toolbox, nav2, and MAVLink

Automated Buckling Load Determination of Slender Column

Electro-Mechanical System Design & Practice

Nov 2023 – Mar 2024

[Poster](#) | [Overview \(Video\)](#) | [Guide \(Video\)](#)

- Designed and developed an automated *electro-mechanical* apparatus for real-time determination of the critical buckling load of slender columns.
- Implemented Arduino Mega-based control, hydraulic actuation, and a real-time buckling detection algorithm for automated testing.
- Technologies: Arduino, Load cell, HX711 ADC, Hydraulic cylinder, MOSFET, Solenoid Valve, and LCD.

Compact Heat Exchanger for Two-wheeler Vehicles

Thermo Fluid System Design

Sep 2024 – Mar 2025

[Report \(PDF\)](#)

- Designed and manufactured a compact heat exchanger featuring flat tubes and triangular louvered fins, achieving a higher heat transfer rate than conventional designs.
- Implemented a Python script to determine the optimal fin and tube geometry, reducing overall weight while maintaining the desired thermal performance.
- Processes: Laser Cutting, Sheet Metal Bending, Welding and Hydraulic Testing.

EXTRACURRICULAR ACTIVITIES

Competitive Programming

Online and Onsite Contests and Problem Solving

- Competed in programming contests from the school level through university, including IOI selection contests, IUPCs, and the ICPC.
- Achieved the **Expert** title (Max Rating: 1739) on Codeforces, demonstrating strong algorithmic problem-solving and competitive programming skills.
- Solved 500+ problems across platforms, covering a wide range of data structures, algorithms, and difficulty levels.
- Best Result: Secured 40th place (*Honorable Mention*) at the ICPC Dhaka Regional, competing against top university teams.

Game Development

Unity | Godot

- Developed multiple projects using Unity 3D, including Android game development, gaining hands-on experience in gameplay programming and engine workflows.
- Contributed to a Unity-based BRTA driver training simulator for manual transmission vehicles, integrating Logitech steering wheel and gear shifter peripherals. | [Demo \(Video\)](#)
- Received the Best BUET Junior Team award in a competitive 96-hour game jam involving over 30 teams across online and on-site phases.

President

2019 – 2020

Josephite Math Club

- Organized the club's flagship event, Josephite Math Mania, managing planning, coordination, and execution.
- Coordinated seminars and academic events while leading member collaboration and task delegation.
- Managed the publication of the annual club magazine *Summation*, including its layout and visual design. | [\(PDF\)](#)

Class Representative

2025-2026

Bangladesh University of Engineering and Technology

ACHIEVEMENTS & AWARDS

- Secured an Honorable Mention, ranking 40th out of 311 competing teams at the **ICPC Dhaka Regional 2025**
- Winner of the **National Math Olympiad** and participant in National Math Camp 2018
- Awarded BUET Rising Team at **DL SPRINT 2.0**, Bengali Document Layout Analysis Competition, BUET CSE FEST 2023 [Presented Paper \(PDF\)](#)
- Awarded Best BUET Juinor Team at **BUET IEEE CS GAMEJAM 2023** [Demo \(Video\)](#)
- Achieved 5th Place at **AI Contest**, BUET CSE FEST 2022 [GitHub Link](#)

TECHNICAL SKILLS

Languages: Python, C/C++, C#, JavaScript, Java

Frameworks: ROS2, Node.js

Developer Tools: Git, VS Code, Visual Studio, PyCharm, Jupyter, Docker, SolidWorks

Libraries: React, pandas, NumPy, Matplotlib

Game Engines: Unity, Godot

Creative Tools: Photoshop, Illustrator, Premiere Pro

REFERENCES

Dr. Mohammad Mamun

Professor

Department of Mechanical Engineering

Bangladesh University of Engineering and Technology

Email: dmmamun@me.buet.ac.bd

Relationship: Thesis Supervisor

Dipak Kumar Sarkar

Assistant Professor

Mathematics

Saint Joseph Higher Secondary School

Email: dsarkarju2981@gmail.com

Relationship: Moderator, Josephite Math Club